

变量分离方程

形如 $\frac{dy}{dx} = f(x)g(y)$ 的微分方程

$g(y) = 0$ 给出特解

$$g(y) \neq 0 \quad \frac{dy}{g(y)} = f(x) dx$$

$$\Rightarrow \int \frac{dy}{g(y)} = \int f(x) dx$$

解 $\frac{dy}{dx} = \frac{x}{y}$

$$y dy = x dx$$

$$\frac{1}{2} y^2 = \frac{1}{2} x^2 + \frac{C}{2} \Rightarrow y^2 = x^2 + C$$

解 $\frac{dy}{dx} = -\cos^2 y \sin x$

当 $\cos^2 y \neq 0$ 时 $\frac{1}{\cos^2 y} dy = -\sin x dx$
 $\tan y = \cos x + C$

当 $\cos y = 0$ 时 当 $y = k\pi + \frac{\pi}{2} \quad k \in \mathbb{Z}$ 时 满足
故有奇解 $y = k\pi + \frac{\pi}{2}$

可化为可分离变量

$$\frac{dy}{dx} = f(ax+by+c)$$

$$\text{令 } u = ax+by+c \quad \text{则 } du = a dx + b dy$$

$$\frac{du}{dx} = a + b \frac{dy}{dx} = a + b f(u)$$

$$\int \frac{du}{a+b f(u)} = \int dx$$

$$\text{解 } \frac{dy}{dx} = (x+2y+1)^2$$

$$\text{设 } u = x+2y+1 \quad \frac{du}{dx} = 1+2u^2$$

$$\frac{du}{1+2u^2} = dx \Rightarrow x = \frac{1}{\sqrt{2}} \arctan(\sqrt{2}u) + C$$

$$\Rightarrow \sqrt{2}u = \tan(\sqrt{2}x+C)$$

$$\Rightarrow y = \frac{1}{2}(u-x-1) = \frac{1}{2} \left(\frac{1}{\sqrt{2}} \tan(\sqrt{2}x+C) - x - 1 \right)$$

$$\text{解 } \frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

$$u = \frac{y}{x}$$

$$f(u) = \frac{dy}{dx} = u + x \frac{du}{dx}$$

齐次

解. $\frac{dy}{dx} = \frac{y^2}{xy+x^2}$

$= \frac{(\frac{y}{x})^2}{\frac{y}{x}+1}$ 设 $u = \frac{y}{x}$, $y = ux$

$\Rightarrow \frac{dy}{dx} = u + x \frac{du}{dx} = \frac{u^2}{u+1}$

$\Rightarrow x \frac{du}{dx} = \frac{-u}{u+1}$

$(1 + \frac{1}{u}) du = -\frac{1}{x} dx \quad u \neq 0$

$\Rightarrow u + \ln|u| = -\ln|x| + \ln|c|$

$y = C e^{-\frac{y}{x}}$

解 $\frac{dy}{dx} = f\left(\frac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}\right)$ 的微分方程

法一:

① $c_1 = c_2$ 时 齐次

② $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ 令 $u = a_1x + b_1y$

③ 设 $u = a_1x + b_1y + c_1$
 $v = a_2x + b_2y + c_2$
 $\Rightarrow dx = \frac{b_2 du - b_1 dv}{a_1 b_2 - a_2 b_1}$
 $dy = \frac{a_1 dv - a_2 du}{a_1 b_2 - a_2 b_1}$

$(a_2 + b_2 f(\frac{u}{v})) du = (a_1 + b_1 f(\frac{u}{v})) dv$

$$\text{法} = \quad \text{令 } x = v + h \quad y = u + k$$

$$\frac{du}{dv} = f\left(\frac{a_1 v + b_1 u + a_1 h + b_1 k + C_1}{a_2 v + b_2 u + a_2 h + b_2 k + C_2}\right)$$

$$\text{令 } \begin{cases} a_1 h + b_1 k + C_1 = 0 \\ a_2 h + b_2 k + C_2 = 0 \end{cases}$$

$$\Rightarrow \frac{du}{dv} = f\left(\frac{a_1 v + b_1 u}{a_2 v + b_2 u}\right)$$

$$\text{解 } \frac{dy}{dx} = \frac{x-y+1}{x+y-1}$$

$$\text{法} - \quad \text{令 } u = x - y + 1 \quad v = x + y - 1$$

$$\text{则 } du = dx - dy, \quad dv = dx + dy$$

$$dx = \frac{du + dv}{2}, \quad dy = \frac{dv - du}{2}$$

$$\Rightarrow \frac{du}{dv} = \frac{1 - \frac{u}{v}}{1 + \frac{u}{v}} \quad \text{令 } z = \frac{u}{v} \Rightarrow z dv + v dz = du$$

$$\frac{du}{dv} = v \frac{dz}{dv} + z = \frac{1 - \frac{u}{v}}{1 + \frac{u}{v}}$$

$$\Rightarrow -\frac{1+z}{(1+z)^2 - 2} dz = \frac{1}{v} dv$$

$$\text{两侧积分} \Rightarrow -\frac{1}{2} \ln |(z+1)^2 - 2| = \ln |v| + C_1$$

$$\text{代入 } x^2 - y^2 - 2xy + 2x + 2y = C$$

$$\frac{dv - du}{dv + du} = \frac{u}{v} \Rightarrow v dv - v du = u du + u dv$$

$$\Rightarrow d\left(\frac{1}{2}v^2 - \frac{1}{2}u^2 - uv\right) = 0$$

$$v^2 - u^2 - 2uv = 0$$

$$\text{法} = : \quad \begin{cases} x = v+h \\ y = u+k \end{cases}$$

$$\frac{du}{dv} = \frac{v+h-u-k+1}{v+h+u+k-1}$$

$$\text{令} \begin{cases} h-k+1=0 \\ h+k-1=0 \end{cases} \Rightarrow \begin{cases} h=0 \\ k=1 \end{cases}$$

$$\Rightarrow \frac{du}{dv} = \frac{v-u}{v+u} \quad \text{齐次}$$

$$\frac{dy}{dx} = \frac{2x+y+1}{4x+2y-3}$$

$$\text{令 } u = 2x+y$$

$$\frac{dy}{dx} = \frac{du}{dx} - 2 = \frac{u+1}{2u-3}$$

$$\frac{du}{dx} = \frac{5u-5}{2u-3} \Rightarrow \frac{2u-3}{u-1} du = 5dx \quad (\text{特解 } u=1)$$

$$2u - \ln|u-1| = 5x + C \Rightarrow \frac{e^{2u}}{u-1} = \underset{\substack{\uparrow \\ \text{令 } \pm}}{C'} e^{5x}$$

$$e^{2(2x+y)} = C' \cdot e^{5x} (2x+y-1) \Rightarrow C \cdot e^{2y-x} = 2x+y-1$$

$$\text{特解 } y = -2x+1$$

